



*Exploring the World of Science*

**2026-7**

## **Florida Science Olympiad Elementary Rules Manual**

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# GENERAL INFORMATION

Science Olympiad's goal is to provide students with a quality competition that links collaboration, inquiry, content understanding, and assessment. Science Olympiad has been highlighted in the 2007 National Governors Report as a national model for learning science and mathematics. Each year, we ensure that we select events that correlate well with Florida's current State Standards.

In the tradition of sporting events, Science Olympiad awards medals/ribbons to the top performing students in each event and trophies to the top performing teams. Students take pride not only in their performance but also in their contribution to the team's performance. Often, students depart the competition with a strong sense of accomplishment, looking forward to next year, yet their ranking may indicate a quite different response would have been expected! Science Olympiad builds a love of science and learning that goes beyond all other academic programs.

This rich experience provides students with something that no other competition does: It encourages teamwork, problem solving, and critical thinking.

## **TEAM SPIRIT:**

Although some events in the Science Olympiad are based on individual achievement, all events involve teamwork, group planning and cooperation. That is the real essence of the Science Olympiad. Our emphasis is on advanced learning in science through active, hands-on, group participation. Through the Olympiad, students, teachers, coaches, principals, business leaders, and parents are all bonded together as a team working toward a goal.

We would like to provide an alternative to the "isolated scientist" stereotype and remind students that science can be fun, exciting and challenging all at the same time. In college and beyond, students will find that the team spirit and good sportsmanship they develop during the Science Olympiad competition will be deciding factors in their success.

## **COACH PREPARATION:**

The most important thing to remember is that this competition is for the students. It is strongly encouraged that the coach takes a facilitation role and accepts the fact that s/he cannot know everything about every event- put the onus on the student to do research and critically read the rules. The events change slightly annually so even long-term coaches will be faced with new challenges every year, which keeps the level of competition high. Many events require students to explain their design choices and/or complete lab work on their own. Teams who have a coach who takes a student-focused approach will have a stronger competitive team.

## **AREA TOURNAMENTS:**

Each elementary level team of 15 will prepare throughout the school year to compete in a Science Olympiad tournament. These inter-scholastic competitions consist of a series of approximately 15 team events that encourage learning in life science, earth science, chemistry and physics, scientific inquiry, technology, and engineering/design.

Events in the Science Olympiad have been designed to recognize the wide variety of skills that students possess. While some events require knowledge of scientific facts and concepts, others rely on science processes, skills, or applications. This ensures that everyone can participate, including students from technology classes or advanced science classes.

Teams are welcome to compete in as many tournaments across the state as desired. However, the team must register and pay the \$180 (\$210) fee for each tournament.

Once you have registered your team, communication for that event will come from the tournament director. Information such as team number may vary from tournament to tournament.

Individual tournaments reserve the right to make rules clarifications or to set their schedule as needed. While most tournaments will run all the events listed in the rules manual, please be sure to see your specific tournament's schedule to confirm.

### **GENERAL EVENT INFORMATION:**

The following apply to all events, even if not noted in the specific rules. Please read carefully:

1. For every event, **it is the responsibility of the competitors to bring their own writing utensils** along with any other items listed in the rules. Teams without eye protection will not be able to compete in any events that require eye protection. Please check the rules for details.
2. **Impound** means that some of your devices or boxes must be put in a holding area before the event begins. These must be turned into specific places at specific times. Anyone on the team can impound the device; parents and coaches are welcome to help the kids with this piece. The team will be penalized if this is not done on time. Generally, impound happens from 8:00 - 8:50.
3. In any open events where spectators are allowed, it is up to the discretion of the event supervisor as to the area where spectators are allowed. The event supervisor has the final right to ask spectators to leave and/or to disqualify teams for members (student, parent, coach, etc.) who are disrupting the flow of the competition.
4. Events are designed to allow for a range of results- some teams will find the event very difficult and some might find it easier. Since this is a competition, events are written for a wide range of final scores and to avoid ties in scores.
5. Chemistry/lab events require specific dress. Please check the manual to ensure that your students will be prepared to compete on the day of the tournament.

**DIFFICULTY OF EVENTS:** This is a competition, so expect that there will be questions on tests and tasks in labs that students may be unable to answer- competitors will be ranked and, therefore, we need a wide range of scores. Work with your students to understand that this is not a situation where they will know everything, and they should do their best. Tests/Labs are generally written at a 4<sup>th</sup>/5<sup>th</sup> grade level with a few questions below that level and a few questions above that level.

### **PARENT/CHAPERONE ROLE DURING COMPETITION :**

1. Parents and chaperones may feel free to move about common areas as identified by the tournament director, but may not interfere with their activities or enter rooms when doors are closed.
2. Parents/chaperones may not yell out instructions to students during open events. Parents should not argue or discuss scoring with the judges. If a judge believes a parent has behaved in a disruptive manner, **this will be grounds for immediate disqualification of the entire team.** STUDENTS may arbitrate judging concerns using the arbitration process.
3. Parents and chaperones should not interact/argue with event supervisors/judges. There is an arbitration process that the STUDENT and coach should complete when concerns arise.
4. If a judge believes a parent has interfered or helped a student in any way, **this will be grounds for immediate disqualification of the entire team.**

### **COACH ROLE DURING COMPETITION :**

Students will be in various rooms around campus. Feel free to move about the common areas as identified by the tournament director, but please do not interfere with their activities or enter rooms when doors are closed. If a judge believes a coach has interfered or helped a student in any way, **this will be grounds for immediate disqualification of the entire team.**

## COMPETITION SCHEDULE:

Schedules may vary from tournament to tournament depending on venue logistics and available supervisors. Teams will follow the predetermined time blocks by team number. Multiple teams from the same school will be scheduled within the same block whenever possible. Below is a **SAMPLE** schedule. It is suggested that tournament directors attempt to follow the sample schedule if possible.

Sample ESO Tournament Schedule for 2027

| DIVISION & EVENTS     | LOCATION | 8 – 8:50  | 9 – 9:50      | 10 – 10:50    | 11 – 11:50 | 12:30 – 1:20 | 1:30 – 2:20 |
|-----------------------|----------|-----------|---------------|---------------|------------|--------------|-------------|
| *AERODYNAMICS         | TBD      | All Build | Set Up        | Self-Schedule |            |              |             |
| A MATTER OF MATTER    | TBD      |           | A01-A08       | A09-A16       | A17-A24    | A25-A32      |             |
| CHEW THE FAT          | TBD      |           | A09-A16       | A17-A24       | A25-A32    | A01-A08      |             |
| CRAB THE WAVE         | TBD      |           | A17-A24       | A25-A32       | A01-A08    | A09-A16      |             |
| CRIME BUSTERS         | TBD      |           | A25-A32       | A01-A08       | A09-A16    | A17-A24      |             |
| DEEP BLUE SEA         | TBD      |           | A01-A08       | A09-A16       | A17-A24    | A25-A32      |             |
| FAST FACTS            | TBD      |           | A09-A16       | A17-A24       | A25-A32    | A01-A08      |             |
| METRIC MASTERY        | TBD      |           | A17-A24       | A25-A32       | A01-A08    | A09-A16      |             |
| MISSION POSSIBLE      | TBD      | Impound   | Self-Schedule |               |            |              |             |
| *MYSTERY PACKAGING    | TBD      | All Build | Set Up        | Self-Schedule |            |              |             |
| PROGAMERS             | TBD      |           | A25-A32       | A01-A08       | A09-A16    | A17-A24      |             |
| ROCKHOUND             | TBD      |           | A01-A08       | A09-A16       | A17-A24    | A25-A32      |             |
| *TENNIS BALL CATAPULT | TBD      | Impound   | Set Up        | Self-Schedule |            |              |             |
| WEATHER PERMITTING    | TBD      |           | A09-A16       | A17-A24       | A25-A32    | A01-A08      |             |
| WRITE IT DO IT        | TBD      |           | A17-A24       | A25-A32       | A01-A08    | A09-A16      |             |
| SPECIAL EVENT         | TBD      |           | A25-A32       | A01-A08       | A09-A16    | A17-A24      |             |

\* Events open for audience based on supervisor's approval. Spectators may NOT interact with participants during testing.

All other events are CLOSED to audience/coaches



## CODE OF CONDUCT :

As part of the paperwork that coaches need to bring to competition, there is a Code of Conduct form to complete by students and the coach. Student participants are expected to compete in tournament events with an honest effort to follow the rules and spirit of the competition. The goal of the competition is to give one's best effort while displaying honesty, integrity, and sportsmanship.

In addition to students, coaches and parents are expected to display courtesy and respect toward Olympiad officials, other teams, and guests of the Olympiad. Please remember all officials are volunteer specialists in their field and should be extended every courtesy.

Failure to show honesty and/or courtesy by a participant, coach, or guest of the team may result in disqualification of the team from that event and/or from the entire competition.

## ARBITRATION/APPEALS:

At times, events might be run in a way that students feel violate the rules. While every effort is made to ensure high correlation with the published rules, there are times when errors by the event supervisor or other issues may arise. In an effort to ensure that all student competitors have an outlet to share their concerns with rules violations, students may complete the arbitration process during the competition.

***See a sample Arbitration form in the Coach Forms section.***

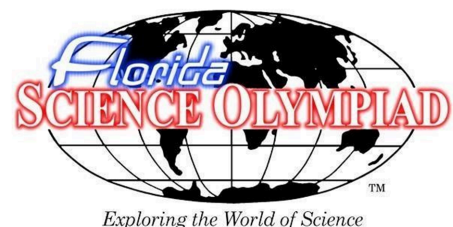
Here are a few notes:

- An arbitration form is found in the coach's envelope and more copies are available at the Help Desk.
- Arbitration forms MUST be completed within ONE HOUR of the end of the event block time where the issue occurred. If necessary, another team member may complete the arbitration process for their team mate if their schedule prohibits this.

- Students may not arbitrate against another team- ex. "The other team did not do ....". Arbitrations are related to the event's alignment with the rules. **These issues should be taken up with the supervisor during the event block in which something happened. We understand that might take a few moments out of your competition time, but these types of issues need to be solved in the event immediately.**
- There will be times when the tournament logistics or limitations of a supervisor will make the event unable to be run in alignment with the rules- we attempt to run all the events even if there may need to be adaptations. In this case, every team will be subject to the same parameters of the event to ensure consistency. Part of Science Olympiad is problem solving and adapting to situations. In these cases, arbitrations will likely end in no change to the procedure.

# FLORIDA ELEMENTARY SCIENCE OLYMPIAD

## EVENT TABLE



### ROTATION OF FUTURE EVENTS :

Below is the tentative rotation of future events. These events may also be substituted for any annual event at the discretion of the tournament director.

Feedback or suggestions for future events is always accepted.

Email [shelley@floridascienceolympiad.org](mailto:shelley@floridascienceolympiad.org) with suggestions/feedback.

|                             | 2027                                                                                             | 2028                                                                                      | 2029                                                                                    | 2030                                                                                       | 3-YEAR ROTATIONS                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Life Science                | Chew the Fat<br>Deep Blue Sea<br>(Freshwater flora and fauna)                                    | A is for Anatomy<br>Backyard Biologist<br>(Trees, insects)                                | A is for Anatomy<br>Backyard Biologist<br>(Trees, Birds)                                | A is for Anatomy<br>Backyard Biologist<br>(Shrubs and insects)                             | A is for Anatomy / Chew the Fat<br>Deep Blue Sea / Backyard Biologist                                                                                         |
| Earth/Space Science         | Rock Hound<br>Weather Permitting                                                                 | Rock Hound<br>Starry, Starry Night                                                        | Fossil Frenzy<br>Starry, Starry Night                                                   | Fossil Frenzy<br>Starry, Starry Night                                                      | Fossil Frenzy / Rock Hound<br>Starry Night / Weather Permitting                                                                                               |
| Physical Science            | A Matter of Matter<br>Crime Busters<br>Crave the Wave                                            | A Matter of Matter<br>Crime Busters<br>Crave the Wave                                     | Circuit Lab<br>Crime Busters<br>Energy Matters                                          | Circuit Lab<br>Can't Judge a Powder<br>Energy Matters                                      | Circuit Lab / A Matter of Matter<br>Crime Busters / Can't Judge a Powder<br>Energy Matters / Crave the Wave                                                   |
| General Science/<br>Inquiry | Metric Mastery<br>Fast Facts<br>(Scientists/Scientific Processes)<br>Write It Do It<br>ProGamers | Metric Mastery<br>Keyed to Science<br>Experimental Design<br>ProGamers                    | Metric Mastery<br>Keyed to Science<br>Experimental Design<br>ProGamers                  | Metric Mania<br>Keyed to Science<br>Experimental Design<br>Codebusters                     | Metric Mania / Metric Mastery<br>Keyed to Science / Fast Facts<br>Experimental Design / Write It Do It<br>ProGamers / Codebusters                             |
| Engineering Design          | Aerodynamics<br>Tennis Ball Catapult<br>Mission Possible<br>Mystery Packaging                    | Balloon Racers<br>Precision Ping Pong Propulsion<br>Mission Possible<br>Mystery Packaging | Balloon Racers<br>Precision Ping Pong Propulsion<br>Roller Coaster<br>Mystery Packaging | Balloon Racers<br>Precision Ping Pong Propulsion<br>Roller Coaster<br>Mystery Architecture | Balloon racers / Aerodynamics<br>Tennis Ball Catapult / Ping Pong Propulsion<br>Mission Possible / Roller Coaster<br>Mystery Architecture / Mystery Packaging |

# AERODYNAMICS

**DESCRIPTION:** Teams will build on site paper airplanes using provided materials and test airplanes based on flight duration and accuracy through a target. This is an On-Site Build event.

## **COMPETITORS BRING:**

- pencils/writing utensils;
- Scissors (one pair per team)
- Ruler (one per team)

**NUMBER OF PARTICIPANTS: 2**

**IMPOUND: NO (ALL TEAMS BUILD ON-SITE)**

## **THE COMPETITION:**

1. Teams will construct (on-site) paper airplanes designed to meet the goals outlined.
2. All teams will construct at the same time to avoid an unfair advantage to those that test first.
3. All teams will return at a later point in the day (as outlined by the schedule) to test their planes.
4. It is suggested that Event Supervisors set up at least two separate areas to test - one for Part 1 and one for Part 2. This will ensure that teams will have adequate time to test in the testing block.

## **BUILD (20 minutes):**

1. Each team member receives 2 pieces of copy paper, 2 paper clips, and 5 cm of masking tape and has 20 minutes to build their airplanes. Type/size of paper clip and width of tape will be consistent among all teams.
2. Each team member constructs his or her own airplane/s - using at least one sheet of paper. But not more than two, which will bear a resemblance to a winged object. No designs like a helicopter are permitted.
3. Teams will write their team number on each plane.
4. One team member's plane/s will be used in Part 1 with the goal being to have the plane in the air the longest. These planes should be labeled "Part 1".
5. The other team member's plane/s will be used in Part 2 with the goals of accuracy to hitting targets. These planes should be labeled "Part 2".
6. Team members should decide whose plane will be used for each part of the competition prior to testing time.
7. The event supervisor should notify teams of where to leave their planes for use during later testing. This location should be monitored so planes are not touched prior to testing.
8. It is suggested that the event supervisor/judge does not touch student planes and instead gives students instructions on where to place them.
9. Teams must clean up after themselves and may be placed in Tier 2 if the construction area is not clean.
10. Teams should leave their design log with the event supervisor during this time.

## **TESTING Part 1 (Duration of Flight)**

1. The goal of this part is for the airplane to stay in the air the longest amount of time.
2. Each plane is launched by hand by its maker from a predetermined starting point, which is the same for all teams. This starting point will be designated by a 1 meter square taped on the floor.
3. Teams should be prepared to launch within 60 seconds of being called upon.
4. Teams may choose to launch one plane at a time (two launches) or launch both planes at once in a "piggyback fashion" one launch
5. If two planes are used at once (in piggyback fashion), the scored time will be for the plane which is aloft for the longer duration.

6. There is no time provided for a test run - a maximum of two launches are allowed to get a maximum of two scores for aloft time. The greater flight time will count as the score for Part 1.
7. Launching team member must give a warning by calling out “3, 2, 1, Launch” or something similar at the time of launch. Failure to do so may result in the loss of the attempt. If this happens, the Event Supervisor should provide a reminder before the next attempt.
8. The time aloft will cease when:
  - a. A plane becomes lodged in a permanent fixture (light, seat, etc)
  - b. A plane touches the floor.
  - c. The plane ceases moving in air or hits a hanging obstacle.

### **TESTING Part 2 (Accuracy of Flight)**

1. The purpose of this part is for the airplane to fly accurately through targets.
2. The planes used in this part must be different from the planes used in Part 1 and should be made by the other team member than in Part 1.
3. Each plane is launched by hand by its maker from a predetermined starting point, which is the same for all teams. This starting point will be designated by a 1 meter square taped on the floor.
4. Teams should be prepared to launch within 60 seconds of being called upon.
5. Teams may choose to launch one plane at a time (two launches) or launch both planes at once in a “piggyback fashion” (one launch)
6. If two planes are used at once (ie. in a piggyback fashion), the accuracy score will be for the plane that goes through the most targets.
7. There is no time provided for a test run - a maximum of two launches are allowed to get maximum score.
8. Two to three targets made from hoops will be erected. Points will be awarded for each hoop through which the airplane passes during its flight.
  - a. The hoops may be of any size (from hula hoops to ones constructed from flexible PVC) but must be stationary and not held by a person. For example, hoops can be held up using purchased hula hoop holders or holders made from a 2-liter bottle with rocks or sand that is attached to the hoop with PVC.
  - b. The distance from the launch area to the initial hoop must be no greater than 3 meters and the distance to the final hoop should be no greater than 10 meters.

### **DESIGN LOG:**

Prior to the competition, teams should prepare a design log. The design log will be submitted during the Construction Block and should include:

- a. Graphs or data charts compiled from flight practice for each part of the competition, graph title, labeled x- and y- axes and using appropriate units for usage of their practice flights.
- b. The log should be clearly labeled with the team number.

Teams who do not submit a design log will be placed in Tier 2.

### **RELATED FLORIDA STANDARDS:**

- **SC.5.P.13.2** Investigate and describe that the greater the force applied to it, the greater the change in motion of a given object.
- **SC.4.P.12.2** Investigate and describe that the speed of an object is determined by its travels in a unit of time and that objects can move at different speeds.
- **SC.5.P.13.3** Investigate and describe that the more mass an object has, the less effect a given force will have on the object's motion.
- **SC.5.P.13.4** Investigate and explain that when a force is applied to an object but it does not move, it is because another opposing force is being applied by something in the environment so that the forces are balanced.

**SCORING:**

1. High score wins.
2. Scoring as follows:
  - a. Score for Part 1 = Longest flight time in seconds X 10
  - b. Score for Part 2 = 50 points for each target through which the plane passes (up to 3)
  - c. Tiebreaker = Quality of design log

Team# \_\_\_\_\_

School Name: \_\_\_\_\_

## Aerodynamics Scoring Sheet

**BUILD:**

|                                                         |                               | TIER |
|---------------------------------------------------------|-------------------------------|------|
| 1. Outside assistance violation?                        | Yes= Tier 2                   |      |
| 2. Planes made only of allowable materials?             | No= Participation points only |      |
| 3. Planes have the team number and Part 1 or 2 on them? | No= Tier 2                    |      |
| 4.. Building area cleaned up as instructed?             | No = Tier 2                   |      |
| 5. Both team members attend the build?                  | No= Participation points only |      |
| 6. Team impounded a data chart/log?                     | No = Tier 2                   |      |

**TEST:**

|                                                                                                                                                                                                                                  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Part 1: Duration of flight<br>Launch 1: _____ seconds      Launch 2: _____ seconds<br>Score will be 10 times the greater of the two launches.                                                                                    |  |
| Part 2: Points for flying through targets<br>Launch 1: _____ targets      Launch 2: _____ targets<br>Score will be given only for launch with the most accuracy.<br>Teams will earn 50 points per target up to 150 points total. |  |
| Design Log: Used for tiebreak only.                                                                                                                                                                                              |  |
| Total Testing Score=                                                                                                                                                                                                             |  |

**TESTING VIOLATIONS:**

|                                                      |                              |  |
|------------------------------------------------------|------------------------------|--|
| 1. Outside assistance violation?                     | Yes= Tier 2                  |  |
| 2. Design alteration violation during testing phase? | Y= Participation points only |  |
| 3. Launch made outside the designated point?         | Yes= Tier 2                  |  |

Total Testing Score: \_\_\_\_\_ Tier: \_\_\_\_\_ Final Rank: \_\_\_\_\_

PLEASE NOTE THAT THIS RUBRIC IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES, THE  
RULES TAKE PRECEDENCE OVER THE RUBRIC.

# A MATTER OF MATTER

**DESCRIPTION:** Students will be asked questions or will conduct experiments at stations as they relate to the properties of matter. (Safety Goggles are required.)

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

## **COMPETITORS BRING:**

- Writing utensil;
- Safety goggles
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

## **THE COMPETITION:**

1. Each team will move from one station to another. There will be at least 4 stations.
2. Team will either be asked questions or will be required to complete experiments or make observations as they relate to the properties of matter.
3. The topics that may be tested or in which an experiment may be conducted include but are not limited to:
  - a. Compare and contrast the physical properties of matter (solids, liquids, and gases) such as mass, shape, volume, color, texture, temperature, attraction to magnets, and hardness
  - b. Describe the changes water undergoes when it changes state through heating and cooling (such as melting, freezing, boiling, evaporation, and condensation)
  - c. Describe the differences between mixtures, solutions, and compounds
  - d. Identify solutes, solvents, and solutions from given examples
  - e. Describe or be asked to separate a given set of mixtures, or solutions using materials provided by the event supervisor
  - f. Determine what will affect the rate of a solid dissolving in a liquid, including temperature, concentration, agitation rate, particle size, that can be tested in a lab. Pressure will not be tested.
  - g. Identify physical versus chemical changes or conduct an experiment to identify and measure the changes
4. Teams may be asked to do experiments dealing with sink/float, changes to states of matter, mixtures/solutions, dissolving rates, etc

## **RELATED FLORIDA STANDARDS:**

- SC.5.P.8.1 Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature
- SC.3.P.8.1 Measure and compare temperatures of various samples of solids and liquids
- SC.3.P.8.2 Measure and compare the mass and volume of solids and liquids
- SC.3.P.8.3 Compare materials and objects according to properties such as size, shape, color, texture, and hardness
- SC.4.P.8.1 Measure and compare objects and materials based on their physical properties including mass, shape, volume, color, hardness, texture, odor, taste, attraction to magnets.

- SC.5.P.8.3 Demonstrate and explain that mixtures of solids can be separated based on observable properties of their parts such as particle size, shape, color, and magnetic attraction
- SC.5.P.8.2 Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process
- SC.5.P.9.1 Investigate and describe that many physical and chemical changes are affected by temperature
- SC.3.P.9.1 Describe the changes water undergoes when it changes state through heating and cooling by using familiar scientific terms such as melting, freezing, boiling, evaporation and condensation
- SC.4.9.1 Identify some familiar changes in materials that result in other materials with different characteristics, such as decaying animal or plant matter, burning, rusting, and cooking

#### **SCORING:**

1. High score wins
2. Each question or station will be assigned a predetermined set of points by the event supervisor and communicated to teams
3. Ties will be broken by predetermined tie-breaker questions which are communicated to teams

# CHEW THE FAT

**DESCRIPTION:** Teams will demonstrate knowledge of the human digestive system and proper nutrition.

**COMPETITORS BRING:**

- Pencil/writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

**NUMBER OF PARTICIPANTS:** 2

**IMPOUND:** NO

**THE COMPETITION:**

This event will be run in a station format. Teams will rotate through stations that assess any or all of the following topics:

1. Identify the major organs and body parts involved in the digestive process and understand the important job each body part has:
  - a. salivary glands
  - b. taste buds
  - c. teeth: incisors, premolars, molars, and canines
  - d. esophagus, stomach, liver, small intestine, large intestine
  - e. kidney, urine, bladder
2. Compare and contrast chemical and physical digestion.
3. Understand food and the benefits of vitamins & minerals.
  - a. Vitamin List: A, B, C, D, E, K
  - b. Minerals List: Calcium, Iron, Magnesium, Phosphorous, Sodium
4. Be able to test for the presence of different nutritional components in foods. For example:
  - a. Fats and oils using the paper bag test and to discover which foods contain high amounts of fats and oils.
  - b. Iron using a magnet in ground up breakfast cereals
  - c. Starch in foods using the iodine test
5. Know how to use food labels to make better food choices, specifically identifying serving size, vitamin and mineral content, overall caloric content plus identifying where the calories are coming from (fat, protein, carbohydrate).
6. Know common foods in the major food groups: fruits, vegetables, grains, protein, dairy and where they fit in the food pyramid and in the USDA My Food Plate.
7. Understand the relationship among the amount of food energy (calories) consumed, weight, and metabolism.

**RELATED FLORIDA STANDARDS:**

- SC.4.N.1.4 Attempt reasonable answers to scientific questions and cite evidence in support.
- SC.4.N.1.6 Keep records that describe observations made, carefully distinguishing actual observations from ideas and inferences about the observations.
- SC.5.N.1.1 Define a problem, use appropriate reference materials to support scientific understanding, plan and carry out scientific investigations of various types.
- SC.5.L.14.1 Identify organs in the human body and describe their functions, including the skin, brain, heart, lungs, liver, intestines, pancreas, muscles and skeleton, reproductive organs, kidneys, bladder, and sensory organs.

**SCORING:** High score wins. Points will be awarded for the accuracy of responses. Ties will be broken by the accuracy or quality of responses to pre-selected questions chosen by the event supervisor.

**RESOURCES:**

[MyPlate.gov](https://myplate.gov) | [U.S. Department of Agriculture](https://www.usda.gov)

<https://www.nutrition.gov/topics/nutrition-life-stage/children/kids-corner>

# CRAVE THE WAVE

**DESCRIPTION:** In this event competitors must demonstrate knowledge and process skills needed to solve problems and answer questions regarding all types of waves and wave motion.

## **COMPETITORS BRING/NEED:**

- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.
- Protractor
- Ruler
- 2 calculators

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

## **THE COMPETITION:**

1. The competition must consist of both hands-on tasks and written questions related to waves.
  - A. 25-50% of the score must be from the practical portion (hands-on tasks)
  - B. 50-75% must be from the theoretical portion (written questions)
2. The event supervisor may provide some mathematical relationships, but the competitors are expected to demonstrate an understanding of the concepts outlined below.
3. The competition must consist of at least one task/question from each of the following areas:
  - A. General wave characteristics (e.g., wavelength, amplitude, frequency, period)
  - B. Wave types (e.g., transverse, longitudinal, surface, torsional)
  - C. Earthquake/seismic waves (e.g., p-waves, s-waves, Rayleigh waves, Love waves, surface waves)
  - D. Illustrating that the sun's energy arrives as radiation with a wide range of wavelengths, including infrared, visible, and ultraviolet, and that white light is made up of a spectrum of many different colors
  - E. Observing and explaining that light and sound waves can be reflected, refracted, and/or absorbed.
  - F. Electromagnetic spectrum (e.g. order, usage, visible light)

## **SCORING:**

- **High score wins.**
- Each question or station will be assigned a predetermined set of points by the event supervisor and communicated to the teams.
- Ties will be broken with pre-determined tie-breaker questions which are communicated to teams.

# CRIME BUSTERS

**DESCRIPTION:** A team of two students will be asked to solve a “crime” by identifying common household white powders, evaluating fingerprints, and completing paper chromatography.

## **SAFETY PARAMETERS:**

Safety Precautions MUST be used:

- Participants dress for the event including closed-toed shoes, long pants, and long sleeve shirt
- Participants must wear safety goggles and lab coat or apron (lab coat may replace long sleeve shirt)
- Shoulder length or longer hair must be tied back
- Taste testing is strictly prohibited

## **COMPETITORS BRING/NEED:**

- Pencils/Pens
- Powders Properties Chart
- Goggles
- Lab Coat or Apron
- Closed-toed shoes must be worn
- Gloves are optional

**NUMBER OF PARTICIPANTS:** Maximum of 3

**IMPOUND:** NO

## **THE COMPETITION:**

1. Students will be given a scenario and clues to solve a “crime”
2. There will be 4 suspects with associated evidence to analyze.
  - a. The competition has 4 parts and may be run as stations:
    - i. Powder Identification: Teams may be asked to identify the following powders in order to solve a simulated crime. Powders will be limited to granulated sugar, flour, cornstarch, baking soda, calcium carbonate (chalk), powdered Alka Seltzer®, plaster of paris, and sand.
      1. Event supervisor will provide powder samples from the “crime”, popsicle sticks or scoops for stirring and scooping powders, hand lens, paper towels, iodine for starch test, white vinegar, droppers or transfer pipettes for liquids, distilled water, and container to test powders in (ex. Well plate, plastic containers, plastic plate)
      2. Participants will create a Powder Properties chart describing the powders’ reactions to the above materials before attending the tournament. 2 copies of this chart should be brought to the tournament. One copy of the chart and answer sheet will be collected at the end of the event.
      3. The samples of powder will be one single powder and not a mixture.
      4. There will be only one powder per suspect to analyze.
      5. Tasting, touching or feeling of the substances or unsafe handling of the equipment is NOT allowed.
      6. No extra powder is provided, so students should practice with small amounts.
    - ii. Fingerprint Identification: Teams will be given a set of fingerprints from several suspects and from the “crime” scene. They will be asked to match the fingerprints found at the scene and identify types of fingerprint patterns such as loops, arches, and whorls.

- iii. Paper Chromatography: Teams will need to perform simple chromatography on markers/pens belonging to suspects in order to match the evidence from the “crime” scene. There will be no more than 1 sample from each suspect. Students will be provided with water, beakers or plastic cups, chromatography or filter paper strips, and various markers/pens to match the evidence from the “crime” scene. No calculations will be performed.
- iv. Case Report: Students analyze the data collected from the three types of evidence and make a conclusion as to who committed the crime and the evidence that supports their conclusion.
- b. Teams will clean stations and dispose of waste as directed by the event supervisor.
- c. Teams must leave their Powders Properties Chart with the event supervisor. The quality of the chart will be used as the first tie-breaker.

#### **RELATED FLORIDA STANDARDS:**

- SC.5.N.1.1 Define a problem, use appropriate reference materials to support scientific understanding, plan and carry out scientific investigations of various types such as: systematic observations, experiments requiring the identification of variable, collecting and organizing data, interpreting data in charts, tables, and graphics; analyze information; make predictions; and defend conclusions.
- SC.4.N.1.6 Keep records that describe observations made, carefully distinguishing actual observations from ideas and inferences about the observations.
- SC.3.N.1.3 Keep records as appropriate, such as pictorial, written, or simple charts and graphs, of investigations conducted.
- SC.3.N.1.6 Infer based on observation
- SC.4.N.1.4 Attempt reasonable answers to scientific questions and cite evidence in support.
- SC.5.P.8.1 Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature.
- SC.5.P.8.2 Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process.

#### **SCORING:**

- High Score wins
- The score will be based on the following formula:
  - Powder identification - 50 points
  - Fingerprint analysis - 20 points
  - Chromotography - 20 points
  - Final Analysis - 25 points
- Tie breakers will be:
  - First tie-breaker - Quality of Powder Properties Table
  - Second tie-breaker - Quality of Final Analysis
- Teams who do not have proper safety equipment cannot compete. They may leave to find someone to lend them the equipment but will not receive any time back to make up for lost time. The Event Supervisor may provide safety equipment, but there will be a 10% penalty given. This penalty is taken off the overall score that the team gets.
- A penalty of 10% will be given if the lab station is not cleaned up as instructed by the event supervisor. This penalty is taken off the overall score that the team gets.

Crime Busters SAMPLE Answer Sheet (May be adapted by Event Supervisor)

School: \_\_\_\_\_ Team Number: \_\_\_\_\_

Team Member 1: \_\_\_\_\_ Team Member 2: \_\_\_\_\_

|                   |           |
|-------------------|-----------|
|                   |           |
| Points for Part 1 | _____/50  |
| Points for Part 2 | _____/20  |
| Points for Part 3 | _____/20  |
| Points for Part 4 | _____/25  |
| Total             | _____/115 |
| Penalty deduction | _____     |
| Final Score       | _____/115 |
| Tiebreaker?       | _____/10  |

Notes/Data Collection:

[illegible]

Instructions: Complete the analysis using observations - use these sheets for answers. No additional paper allowed.

Part 1: Powder Evidence:

|                            | Evidence                                                                     | Analysis of the Evidence (Answer) | Score  |
|----------------------------|------------------------------------------------------------------------------|-----------------------------------|--------|
| 1a                         | What is the identity of the powder found on Suspect 1?                       |                                   | ___/4  |
| 1b                         | How did you identify it?                                                     |                                   | ___/5  |
| 2a                         | What is the identity of the powder found on Suspect 2?                       |                                   | ___/4  |
| 2b                         | How did you identify it?                                                     |                                   | ___/5  |
| 3a                         | What is the identity of the powder found on Suspect 3?                       |                                   | ___/4  |
| 3b                         | How did you identify it?                                                     |                                   | ___/5  |
| 4a                         | What is the identity of the powder found on Suspect 4?                       |                                   | ___/4  |
| 4b                         | How did you identify it?                                                     |                                   | ___/5  |
| 5a                         | What happened when you tested the evidence powder found at the scene?        |                                   | ___/5  |
| 5b                         | What powder was found at the scene?                                          |                                   | ___/5  |
| 5c                         | Which suspect's mystery powder is the same as the powder at the crime scene? |                                   | ___/4  |
| Total points for Section 1 |                                                                              |                                   | ___/50 |

Part 2: Fingerprint Analysis:

|    | Evidence                                 | Analysis of the Evidence (Answer) | Score |
|----|------------------------------------------|-----------------------------------|-------|
| 1a | Type of fingerprint pattern of suspect 1 |                                   | ___/3 |
| 1b | Type of fingerprint pattern of suspect 1 |                                   | ___/3 |

|                            |                                                       |  |        |
|----------------------------|-------------------------------------------------------|--|--------|
|                            |                                                       |  |        |
| 1c                         | Type of fingerprint pattern of suspect 3              |  | ___/3  |
| 1d                         | Type of fingerprint pattern of suspect 4              |  | ___/3  |
| 1e                         | Type of fingerprints pattern found at the crime scene |  | ___/3  |
| 1f                         | Whose fingerprints were like those at the scene?      |  | ___/5  |
| Total points for Section 2 |                                                       |  | ___/20 |

### Part 3: Paper Chromatography:

Complete a chromatograph according to the directions at the lab station.

|                            |                                                                                                      |  |        |
|----------------------------|------------------------------------------------------------------------------------------------------|--|--------|
| 3a                         | Paperclip or staple your chromatography below. Points will be awarded for the quality of separation. |  | ___/15 |
| 3f                         | Points awarded for:<br>Marking the baseline (2.5)<br>Marking the solvent front (2.5)                 |  | ___/5  |
| Total points for Section 3 |                                                                                                      |  | ___/20 |

### Part 4: Case Report:

Instructions: Using the evidence you analyzed, draw a conclusion as to what happened about the event. Please consider evidence that supports the guilt of certain suspect(s) and evidence that supports the innocence of the other suspects. This section will be used as the second tiebreaker, if needed.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

# DEEP BLUE SEA (Freshwater Flora and Fauna)

**DESCRIPTION:** Teams will be assessed on their knowledge of oceanography, the freshwater flora and fauna of Florida, and aquatic animal life cycles.

## **COMPETITORS BRING/NEED:**

- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

## **THE COMPETITION:**

1. The competition may include timed stations and/or timed slides/PowerPoint presentation(s).
2. In **Part I** the contestants will view pictures and answer questions relating to identifying **Intracoastal Waterway flora or fauna**. They may also be asked questions about the **Intracoastal Waterway organism's habitat, diet, or life cycle**.
3. The list of flora and fauna will come from the following list:

### **FLORA:**

|               |                            |               |
|---------------|----------------------------|---------------|
| Sawgrass      | Widgeon grass              | Sweetgum      |
| Water lily    | Bald cypress               | Buttonush     |
| Cattail       | Pond cypress               | Yaupon        |
| Maidencane    | Water tupeloSwamp blackgum | Hydrilla      |
| Pickernelweed | Sweetbay magnolia          | Swamp rose    |
| Bladderworts  | Pond apple                 | Cinnamon fern |
| Spikerush     |                            | Royal fern    |

### **FAUNA:**

|                             |                          |                            |
|-----------------------------|--------------------------|----------------------------|
| American Alligator          | Catfish                  | Americana flamingo         |
| Banded Southern Water Snake | Florida Gar              | Crested caracara           |
| Water Moccasin              | Tarpon                   | Osprey                     |
| Florida Softshell Turtle    | Smalltooth Sawfish       | Sandhill crane             |
| Florida SnappingTurtle      | Everglades Pygmy Sunfish | Wood stork                 |
| Diamondback Terrapin        | Largemouth Bass          | Anhinga                    |
| White tailed deer           | Gulf Toadfish            | Double-crested cormorant   |
| Florida panther             | Snook                    | Turkey vulture             |
| Bobcat                      | Peacock Bass             | Limpkin                    |
| North American river otter  | Seminole Killifish       | Peregrine falcon           |
| Marsh rabbit                | Brim                     | Great blue heron           |
|                             | West Indian Manatee      | Woodstork                  |
|                             | Mayfly                   | Redshouldered hawk         |
|                             | Caddisfly                | Snail kite                 |
|                             | Dragonfly                | Cape Sable seaside sparrow |
|                             | Peregrine falcon         |                            |

4. In **Part II** the contestants will respond to a series of questions related to the following topics:
  - a. Physical features of oceans (trenches, seamounts, etc.)
  - b. Phenomena (tidal waves, currents, surface current, longshore currents, etc.)
  - c. Geography (location and identification of oceans, seas, major bays, etc.)
  - d. Conservation efforts to protect ocean animals and sea floor

**RELATED FLORIDA STANDARDS:**

3. SC.3.L.14.1 Describe structures in plants and their roles in food production, support, water, and nutrient transport, and reproduction.
4. SC.3.L.14.2 Investigate and describe how plants respond to stimuli (heat, light, gravity), such as the way plant stems grow toward light and their roots grow downward in response to gravity.
5. SC.3.L.15.1 Classify animals into major groups (mammals, birds, reptiles, amphibians, fish, vertebrates and invertebrates, those having live births and those which lay eggs) according to their physical characteristics and behaviors.
6. SC.4.L.16.4 Compare and contrast the major stages in the life cycles of Florida plants and animals such as those that undergo incomplete and complete metamorphosis, and flowering and nonflowering seed-bearing plants.
7. SC.4.L.17.4 Recognize ways plants and animals, including humans, can impact the environment.
8. SC.5.L.15.1 Describe how, when the environment changes, differences between individuals allow some plants and animals to survive and reproduce while others die or move to new locations.
9. SC.5.L.17.1 Compare and contrast adaptations displayed by animals and plants that enable them to survive in different environments such as life cycles, variation, animal behaviors, and physical characteristics.

**SCORING:**

- High score wins.
- Each question or station will be assigned a predetermined set of points by the event supervisor and communicated to the teams.
- Ties will be broken with pre-determined tie-breaker questions which are communicated to teams.

**RESOURCES:**

# FAST FACTS (Scientists/Scientific Processes)

**DESCRIPTION :** Teams will fill in a grid of terms that begin with a given letter to match given science categories related to scientists and scientific processes..

**COMPETITORS BRING/NEED :**

- Writing utensil

**NUMBER OF PARTICIPANTS :** Maximum of 2

**IMPOUND :** NO

**THE COMPETITION :**

1. Teams are not allowed to bring any information resources or other material into the event.
2. The event supervisor will provide each team answer/scoresheets.
3. Each competition will consist of four rounds. Each round will begin with the supervisor giving each team the same scoresheet that contains a grid which has 5 different science categories listed along the horizontal axis and 5 different letters listed along the vertical axis. The supervisor will determine the categories to be used in each round, ensuring that there is at least one valid answer for each category/letter combination. Categories must not be repeated within a grid or among the four grids used in competition. No letter can be repeated more than twice within the four grids.
4. Teams will have 6 minutes to complete each round. Teams will write a term, corresponding to the given category and beginning with the given letter, in each of the 25 boxes of the grid. At the end of 6 minutes the event supervisor will stop the round. For each round all students should start and stop writing at the same time as directed by the event supervisor. Students beginning before or working after the event supervisor starts/stops the round will have their scoresheet not scored for that round.
5. Students are to write their names and team number on the scoresheet for each round. A scoresheet without student names and/or school will not be scored for that round.
6. At the end of each round the supervisor will collect all scoresheets. Then a new set of scoresheets will be distributed to students. This will repeat for each of the three rounds.
7. Names of the categories must not be used in the answer.
8. If a correct response has more than one word, the first letter of the first word will be used (e.g., "D" is the first letter of "Doppler Radar"), Exceptions: the first letter of a word following articles, "the", "a/an" will be considered the first letter of the term (e.g., "The Grand Canyon" would be considered to begin with "G".)
9. Students may not write two or more different forms of a response within a category to get credit for two or more different answers (e.g., Category - "Human Organs", Letters "I", "L", "S", and the student writes "small intestine", "large intestine", and "intestine". The student would only get credit for the more precise terms "small intestine" and "large intestine".
10. Incorrect spelling of words will be allowed if the event supervisor is able to determine the intended term. However, the first letter must be correct. All words must be found in an English based science dictionary.

**RELATED FLORIDA STANDARDS :**

- SC.3-5.E.5.1-5 - this includes and will not be limited to all elementary standards on Earth/Space Science

**SCORING :**

- Total score will be determined by adding the scores from each row and column from each of the four rounds. Highest total score wins.
- The number of points earned depends upon the number of correct terms listed in a row and in a column. Points will be awarded as follows:
  - One correct term in a row = 1 point
  - Two correct terms in a row = 4 points

- Three correct terms in a row = 9 points
- Four correct terms in a row = 16 points
- Five correct terms in a row = 25 points
- One correct term in a column = 1 point
- Two correct terms in a column = 4 points
- Three correct terms in a column = 9 points
- Four correct terms in a column = 16 points
- Five correct terms in a column = 25 points
- Ties will be broken in the following sequence
  - Highest individual round score
  - Second highest individual round score
  - Most columns/rows with 5 correct answers
  - Most columns/rows with 4 correct answers
  - Most columns/rows with 3 correct answers
  - Most columns/rows with 2 correct answers

**RELATED FLORIDA STANDARDS:**

10. SC.3-5 E.5.1-5 - this includes all elementary standards on Earth/Space Science

# METRIC MASTERY

**DESCRIPTION :** Teams will estimate and then measure properties of identical objects including mass, area, volume, distance, time, etc. Teams will also perform metric unit conversions.

**COMPETITORS BRING/NEED :**

- Writing utensil
- Simple calculator

**NUMBER OF PARTICIPANTS :** Maximum of 2

**IMPOUND :** NO

**THE COMPETITION :**

1. The event will be divided into 3 sections. All measurements and calculation units must be limited to the metric system.
2. Section 1 - Estimation - will involve estimating measurement properties of objects at stations. During Section 1 students will be estimating only and no measurement tools will be used.
3. The property to be estimated and the units of measure must be clearly identified in the directions at the station.
4. Part 1 will be set up as 8 individual stations. Teams will have 60 seconds per station to make their estimates.
5. Participants may not touch or feel the objects unless the station directions clearly state otherwise. (students must be allowed to lift the object if they are being required to estimate mass.)
6. Section 2 - Metric Unit Conversion - Teams will have 10 minutes to complete metric unit conversions.
7. This part must be completed after Part 1.
8. Participants will have 10 minutes to complete metric unit conversion problems to a specific specified metric unit. Conversions must be limited to the metric system only.
9. Section 3 - Measurement - will involve precisely measuring properties of objects at stations. Some of the stations may ask for calculated measurements. When this is the case, formulas will be given.
10. Measurements must be made using the supervisor-supplied instruments, expressed in the units of the instrument and as specified in the directions.
11. To receive points, the units must be given.
12. Part 3 will be a repetition of the stations from Section 1. Teams will have 120 seconds per station to make and calculate their measurements.
13. During Part 3 students will be allowed to carefully touch the objects and the measuring tools provided.

**Estimations/Measurements which may be included:**

- Area - 2D shapes and combined 2D shapes
  - Mass
  - Length/Distance
  - Density
  - Perimeter
  - Angles
  - Time
  - Liquid volume
  - Volume - of rectangular prisms or of other objects using water displacement
  - Surface area of 3D shapes
14. Calculated measurements are limited to area and volume of polygons and polyhedrons, density of contained liquids, solids, and powders, displacement of liquid by a solid, and mass of solids and liquids.

**SCORING:**

1. High Score Wins. Final score = Estimation Score + Conversion Score + Measurement Score
2. Section 1 - Estimation Score (for a total of up to 40 points)

|                                     |                                     |                                    |
|-------------------------------------|-------------------------------------|------------------------------------|
| Answers within 10% receive 5 points | Answers within 20% receive 3 points | Answers within 30% receive 1 point |
|-------------------------------------|-------------------------------------|------------------------------------|

3. Section 2 - Metric Unit Conversion Score (for a total of up to 40 points)  
Answers must have the metric unit included to receive points.

4. Section 3 - Measurement Score (for a total of up to 80 points)

|                                      |                                        |                                        |                                          |                                        |
|--------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|----------------------------------------|
| Answers with 5%<br>receive 10 points | Answers with 10%<br>receive 7.5 points | Answers within 15%<br>receive 5 points | Answers within 20%<br>receive 2.5 points | Answers beyond 20%<br>receive 0 points |
|--------------------------------------|----------------------------------------|----------------------------------------|------------------------------------------|----------------------------------------|

5. Ties will be broken by predetermined stations within Section 3 of the event.

6. Penalties, up to 10%, may be applied to teams who do not return measuring devices to their original locations, do not clean up spills, and/or intentionally alter or damage equipment or objects.

# MISSION POSSIBLE

**DESCRIPTION:** Teams will design, build, test and bring a Rube Goldberg-like device which incorporates up to 12 unique Action Transfers and uses five forms of energy in accomplishing a given task in 60 seconds. Devices will be constructed prior to competition. This is a **PRE-BUILT EVENT**.

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** Yes

## **COMPETITORS BRING:**

- Device
- 2 copies of the Action Sequence List (ASL)
- Safety goggles
- Any materials needed to repair device

## **THE COMPETITION:**

### **CONSTRUCTION:**

#### **1. GENERAL REQUIREMENTS:**

- a. The device must fit inside an imaginary box **75 cm<sup>3</sup>** and must fit through a single door opening. Devices not meeting these requirements will be placed in Tier 2. Device may be on wheels, but must still comply with size requirements
- b. The device may use ambient room light and must not depend on direct sunlight to operate. All other sources of energy and actions must take place within the imaginary box before, during, and after the device's operation.
- c. **All parts, pieces, substances must remain within the device at all times. Any violation will result in a 20 point penalty per offense.**
- d. Students must be able to move the device from one location to another (from impound to testing).
- e. For measured actions, team must have a measurement tool to show the required distance (this can be a portion of a paper centimeter ruler)

#### **2. SCORABLE ACTIONS:**

- a. Start Action: **(40 Points)** - The device will be started by dropping a golf ball from completely above the device to initiate a subsequent action. This should be clearly identified on the ASL for points.
- b. Scorable Actions: **(20 points each for a total of 240)** - each scorable action needs to initiate the next scorable action.
  - i. Use a 3:1 gear system to initiate an action
  - ii. Use a wheel and axle to raise an object at least 10 cm vertically, that object should then initiate an action **(the measurement should be marked on the device)**
  - iii. Transfer gravitational potential energy to kinetic energy to initiate an action
  - iv. Use chemical energy to initiate an action
  - v. Knock over a series of at least 5 free-standing, commercial dominoes, the last of which initiates an action
  - vi. Use a pulley system to raise an object at least 10 cm vertically, that object should then initiate an action **(the measurement should be marked on the device)**
  - vii. Push or pull an object up an inclined plane to initiate an action
  - viii. Use wind energy to initiate an action
  - ix. **Use water to raise an object at least 5 cm to initiate an action (the measurement should be marked on the device)**
  - x. Pull a wedge from underneath an object to initiate an action
  - xi. Blow up a balloon and use that to initiate an action

- xii. Use a wheel and axle to move an object at least 10 cm horizontally, that object should then initiate an action **(the measurement should be marked on the device)**
- xiii. *Actions must be designed to contribute to the final action and should be consecutive. Actions that are parallel or dead-end actions will not count for points.*
- xiv. *Scorable actions cannot be combined and should be identified on the ASL. For example, if you use a funnel to transfer gravitational energy to kinetic energy, the team must identify which of those is the single scorable action. A battery can not count as both chemical and electrical energy if initiating the same action.*
- xv. *Energy sources (i.e. springs, mousetraps) must be set prior to starting the device in the 5 minute maximum set up time.*
- xvi. *Heat energy may be used but must not have a flame.*
- xvii. *No electric device may have or use an electric potential difference of more than 9 volts.*
- xviii. *If electric devices used for heat are rechargeable, documentation must be submitted to confirm output is no more than 9 volts.*
- c. Final Action: **(40 points)** Teams will design the device to complete the final task of **raising a flag completely above the top of the device**, at the end of 60 seconds. There are no specific requirements on the flag, but it should be recognizable as a flag.

### 3. ACTION SEQUENCE LIST (ASL)

- a. Teams will use the template provided to create an Action Sequence List corresponding to their device.  
[w ESO Mission Possible Action Sequence List 26-27.docx](#)
- b. Teams must submit TWO copies of the Action Sequence Lists (ASL) to the Event Supervisor during impound.
  - i. Your ASL must describe the sequence in which the Action Transfers will occur (this must include a brief description, not just a list)
  - ii. Your ASL must indicate the Form of Energy used in each action
  - iii. Your ASL must have your team number clearly marked on it.
- c. The ASL may be kept by the supervisor for use in scoring and/or score counseling.
- d. Link to ASL template. Be sure to save a copy so that you can edit the document.  
[w ESO Mission Possible Action Sequence List 26-27.docx](#)
- e. Teams without an ASL will be placed in Tier 2.

## THE COMPETITION:

### 1. IMPOUND:

- a. All devices must be impounded during the indicated impound time. Teams who do not impound may not be allowed to compete or will be placed in Tier 2 based on the supervisor's discretion.
- b. Adults may be able to help with impound but will not be able to assist during testing blocks. Teams should ensure that students are able to move their device from the impound location to the testing location. If adults enter the testing area, the team may be placed in Tier 3.
- c. Each device must pass a safety inspection BEFORE operation. Uncontrolled or hazardous non-shielded, falling off launch objects, hazardous materials or spills, hazardous flammable substances, faulty wiring, or any other potential hazard can result in being unable to compete and earning only participation points for the event (Tier 3)
- d. Two copies of the ASL will be impounded with the device.
- e. Teams should impound goggles for use during testing.
- f. The team must also impound any materials which may be needed during the testing process or to repair the device. No other materials will be allowed.

### 2. COMPETITION:

- a. The event will be held indoors.

- b. Teams will have a maximum of 5 minutes to move and reset their device for testing. Teams will earn a bonus of 20 points if they start their device within 2 minutes of their competition time.
- c. Within 5 minutes of their competition time, the team will notify the supervisor that they are ready to test. Students must have their goggles on to begin the run. Teams who do not have goggles will be given one warning and allowed to get goggles but their 5 minute time will not be stopped. Teams without goggles will be placed in Tier 2.
- d. The team must follow the directions of the supervisor to start the device.
- e. Teams should not touch their device unless the device stops, jams, or fails. In that case, the participants are allowed to touch/adjust it to continue operation. Participants do not need to wait for the event supervisor's permission but should notify the supervisor of the restart.
  - i. Teams will lose 10 points for each restart needed during the run with a maximum of 3 restarts allowed before time is called on the run.
  - ii. Obvious stalling or adjusting only to impact runtime will result in disqualification (Tier 3).
- f. Teams will be allowed to run their device only one time.
- g. Run time ends when:
  - i. The final task is complete
  - ii. The device has been restarted 3 times and stops (Time will be recorded as 2 minutes and only successful scorable actions will be awarded points)
  - iii. Two minutes have passed
- h. The event supervisor may interview students about their device. Therefore, team members should be prepared to identify scorable actions on their device and identify the forms of energy and energy transformation taking place.

#### **RELATED FLORIDA STANDARDS:**

- **SC.5.P.10.1** Investigate and describe basic forms of energy, including light, heat, sound, electrical, chemical, and mechanical
- **SC.5.P.10.2** Investigate and explain that energy has the ability to cause motion or create change
- **SC.5.P.10.4** Investigate and explain that electrical energy can be transformed into heat, light, and sound energy, as well as energy of motion.

#### **SCORING :**

1. **High score wins**
2. The Action Score will be calculated using the following:
  - a. 40 points for a starting action
  - b. 20 points for each scorable action
  - c. 40 points for final action
3. The runtime score will be 60 points minus 1 point for each second below or beyond the 60 second target time.
4. Bonus points will be awarded for early start as stated above.
5. Penalties will be deducted for restarts and violations for materials outside of device as stated above.
6. Ties will be broken by the quality/accuracy of the ASL.

# MYSTERY PACKAGING

**DESCRIPTION :** At the beginning of the event, teams will be given a bag of building materials and instructions for designing a package that will protect a given material from a drop. This is an ON-SITE BUILD EVENT.

**COMPETITORS BRING/NEED :** (per student)

- Allowable tools (Writing utensil, Scissors, Ruler)
- Safety goggles

**NUMBER OF PARTICIPANTS :** Maximum of 2

**IMPOUND :** NO

**THE COMPETITION :**

**BUILD:**

1. All teams will build at the same time. Teams will have 25-minutes to build. Once teams enter the area to build, they may not leave the area or receive outside assistance, materials or communication until after they are finished building. Violation of this rule places a team in Tier 2.
2. Once participants arrive at their build area, they must put on their safety goggles. These goggles must stay on for the entire build period. Supervisors will give one goggle warning. Additional warnings will place the team in Tier 2 and/or remove them from competition.
3. Each team will be given a bag containing the same materials and working area.
  - a. Examples of materials are : paper cups, drinking straws, paper clips, string, tape, paper, Play Doh, and craft sticks. Materials are not limited to this list but will not include anything potentially hazardous.
  - b. Teams may not use any outside materials, other than what is provided to them by the event supervisor or their allowable materials listed above.
  - c. The bag that the materials come in cannot be part of the package design and should be used to hold all waste at the end of the build.
  - d. The materials provided may be altered as needed, using the allowable tools.
  - e. The allowable tools may not be part of the package.
4. The material that will be loaded into the device will be announced at the start of construction.
  - a. This could be either a solid or a liquid and either homogeneous or heterogeneous in composition.
  - b. After the teams have constructed the package, the material will be loaded into the device during the test phase.
  - c. Teams will be allowed to examine a separate "inspection" load material of the same size and weight as used for testing but this cannot be loaded into the device during construction. This inspection area will be separate from the building area and may be accessed by any team, however teams may not take the sample material back to their building area.
5. The team will create a package using the given materials that will hold the load when it is dropped from a height between 0.5 and 1 meter, depending on the team's choice.
  - a. The device must be built to allow for loading of the material after construction has been completed.
  - b. The loading of the material should be able to be completed and the device ready to drop within one minute.
  - c. The package must have the team number on it. Packages without team numbers will be placed in Tier 2.
6. At the end of the build time, teams must clean up their build area by placing all trash and extra materials in the bag.
  - a. The remaining supplies will be massed and this will be used as a tie-breaker (those with more supplies left over will place higher in the tie.)
  - b. Prior to leaving the build area, teams must ensure that their area is clean and all instructions have been followed. Failure to follow this rule will place teams in Tier 2.

- c. Teams should store their packages in a location noted by the event supervisor. It is suggested that the event supervisor/judges do not touch student packages and instead have them place the packages in a location that will be monitored throughout the day.

#### TEST:

1. Teams will return to test at some point later in the day - this may be self-scheduled or based on a set schedule.
2. Once participants arrive at their test area, supervisors will have them collect their device and give them the loading material (this will be the same for all participants.)
3. Teams have one minute to load the material into their package and balance the package on the scale to measure the mass. Teams who do not meet this time will earn participation points only.
4. Teams whose package cannot hold the entire load will earn participation points only.
5. Teams will then take their full package and proceed to the testing site.
6. One member of the team will hold a meter stick and the other will hold the meterstick and the other will hold the package at the drop height of their choice. This drop height should be between 0.5 and 1 meter. The height is measured from the bottom of the package to the floor. The supervisor will confirm the height.
7. One given the signal by the event supervisor, teams will drop their full package. Teams have one chance to drop their package. There are no trial runs or do-overs.
8. Teams will pick up their package and carry it to the scale to measure the ending mass to account for lost load. Any material that falls out prior to massing will not count in the final mass.
9. Teams will remove the load according to the supervisor's instructions and return their package to the location where they were stored. Teams may pick up their packages at the end of the day, if desired.

#### RELATED FLORIDA STANDARDS:

SC.5.N.1.5 - Recognize and explain that authentic scientific investigation frequently does not parallel the steps of "the scientific method"

SC.5.P.13.1 - Identify familiar forces that cause objects to move, such as pushes or pulls, including gravity acting on falling objects.

#### SCORING:

1. High Score wins.
2. The following formula will be used to get the test score:  
$$\text{Score} = [\text{drop height} - (\text{initial loaded package mass} - \text{final package mass})] \times 100$$
  - Drop height will be measured in centimeters
  - Mass will be measured in grams
  - Negative point values are allowed
3. Devices that cannot support the load will be given participation points only.

Team# \_\_\_\_\_

School Name: \_\_\_\_\_

## Mystery Packaging Scoring Sheet

| BUILD:                                   |                                                                                                       | TIER |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|------|
| Outside assistance violation?            | Y= Tier 2; Repeated violations may result in DQ                                                       |      |
| Safety goggles violation?                | 1 <sup>st</sup> time= Warning; 2 <sup>nd</sup> time = Tier 2;<br>Repeated violations may result in DQ |      |
| Device made only of allowable materials? | N= Participation points only                                                                          |      |
| Device has team number on it?            | N= Tier 2                                                                                             |      |
| Building area cleaned up as instructed?  | N = Tier 2                                                                                            |      |
| Team did not show up at build time?      | Y= Participation points only                                                                          |      |
| TIER?                                    |                                                                                                       |      |

## TEST:

|                                                        |                              |  |
|--------------------------------------------------------|------------------------------|--|
| Mass of bag with "waste"                               | _____ g (TIE-BREAKER)        |  |
| Loaded within 1-minute?                                | N= Participation points only |  |
| Device held the load?                                  | N= Participation points only |  |
| Mass of fully loaded package BEFORE drop in grams      |                              |  |
| Height of drop in centimeters                          |                              |  |
| Mass of package and remaining load AFTER drop in grams |                              |  |
| Change in Mass                                         |                              |  |

|                   |   |                         |   |       |             |
|-------------------|---|-------------------------|---|-------|-------------|
| Drop Height in cm | - | Change in mass in grams | X | 100 = | Final Score |
|                   | - |                         | X | 100 = |             |

PLEASE NOTE THAT THIS SCORESHEET IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES,  
THE RULES TAKE PRECEDENCE OVER THE SCORESHEET.

# PROGAMERS

**DESCRIPTION:** Teams of students will use the latest Scratch programming language to recreate a game being shown to them on a screen in the room.

**COMPETITORS BRING/NEED:**

- Writing utensils
- One sheet of 8.5 X 11 sheet of paper, which may include hand written information on both sides from any source (per team)

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

**THE COMPETITION:**

1. Each team will have their own computer with access to the Scratch website.
2. Students will log in using the accounts created specifically for the tournament.
3. A pre-programmed game, supplied by the event supervisor, will play on a continuous loop on a screen in the room.
4. Students will have 40-45 minutes (all teams will have the same amount of time for the tournament) to program the game exactly as it is shown.
5. Students may be asked to code part of the game using a specific technique (e.g. use a loop)
6. Students will make their game public by sharing them with the event supervisor.

**SCORING:**

- High score wins.
- Points will be award for each piece of the code that students are successfully able to program.
- Ties will be broken with pre-determined pieces of code or by the team that successfully programs a piece of code in the least amount of time. EVent supervisors will determine and communicate the tiebreaker to teams.

# ROCK HOUND

**DESCRIPTION:** Teams will demonstrate knowledge of types and properties of rocks and minerals as well as natural resources.

**COMPETITORS BRING:**

- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

**THE COMPETITION:**

1. The event may be run as stations, may include a slideshow of images, and/or may be a “pencil and paper” test.
2. Teams will identify as many rocks and minerals as possible from a selected group (may be from images or samples). The list of rocks and minerals will come from the following list:

|           |                                                 |                                                             |                                                        |                                                               |
|-----------|-------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------|
| ROCKS:    | basalt<br>granite<br>pumice<br>scoria           | bituminous coal<br>limestone (fossil)<br>quartzite<br>shale | conglomerate<br>marble<br>sandstone<br>slate           | gneiss<br>obsidian<br>schist (garnet)<br>limestone (chemical) |
| MINERALS: | calcite<br>galena<br>Hematite<br>Quartz (chert) | copper<br>graphite<br>mica-biotite<br>quartz (crystal)      | feldspar (pink)<br>gypsum-satin-spar<br>pyrite<br>talc | fluorite<br>halite<br>kaolinite                               |

3. No more than half of the questions on the exam will be identification questions.
4. Additional questions may be from any of the following topics:
  - a. Categories of rocks (igneous, metamorphic, sedimentary)
  - b. The rock cycle
  - c. Weathering and erosion
  - d. Characteristics of minerals (hardness, luster, color, cleavage, streak)
  - e. Natural resources (renewable vs. nonrenewable, conservation of resources, Florida resources such as phosphate and limestone)
5. Students may be asked to perform a streak test.

**RELATED FLORIDA STANDARDS:**

SC.4.E.6.1 Identify the three categories of rocks: igneous, (formed from molten rock); sedimentary (pieces of other rocks and fossilized organisms); and metamorphic (formed from heat and pressure).

SC.4.E.6.2 Identify the physical properties of common earth forming minerals, including hardness, color, luster, cleavage, and streak color, and recognize the role of minerals in the formation of rocks.

SC.4.E.6.3 Recognize that humans need resources found on Earth and that these are either renewable or nonrenewable.

SC.4.E.6.6 Identify resources available in Florida (water, phosphate, oil, limestone, silicon, wind, and solar energy).

**SCORING:**

1. High score wins.
2. Each question or station will be assigned a predetermined set of points, assigned by the event supervisor and communicated to the teams (preferably- points are identified on the test/station paperwork).
3. Ties will be broken with pre-determined tie-breaker questions which are communicated to teams.

# TENNIS BALL CATAPULT

**Description:** Students will build and calibrate their own free-standing (not handheld) trajectory device that must be capable of “lobbing” a tennis ball at a target placed between 5 and 8 meters away. Students MUST bring and wear impact-resistant safety goggles. **This is a PRE-BUILT EVENT.**

**Number of Participants:** 2

**Impound:** YES

## **The Competition:**

1. Each team will bring their own homemade catapult or trebuchet and their own standard unaltered regulation tennis balls to the competition. It is suggested to bring three tennis balls. Print your team number prominently on the catapult and balls. If the team fails to bring their own tennis balls, they may be supplied by the judge at a penalty of +20 points.
2. Launch force must be provided by gravity or elastic solids (such as springs, rubber bands, bungee cords, etc.).
3. The device may not be more than 50 cm above the ground before, during or after launching. The total weight of the device must be under 50 pounds. Students must be able to move their device from the impound area to the launch site without adult assistance. If the device violates these design parameters, the team will be placed in Tier 2
4. Each device should be designed and built by the students (adult construction assistance is OK). Each device should be designed to operate safely at all times. No points will be awarded for design.
5. The device must sit upon rubber or soft material “feet” to prevent floor damage when moving and launching.
6. The device will sit on a level area of ground and fire at the target area that will also be at ground level. The target area will be a freshly raked sand pile of at least 1 meter. A small object will mark the target.
7. The event may take place outside.
8. **The team may not use any digital or electronic devices to aid in aiming or measuring.**
9. Prior to the competition, teams should prepare a design log. The design log will be submitted during impound and should include:
  - a. At least 2 diagrams showing different views of the device and details of construction
  - b. A graph or data chart compiled from launch practice, graph title, labeled x- and y- axes and using appropriate units for usage of their device.
  - c. The log should be clearly labeled with the team number.
  - d. Students should be prepared to answer questions about their device and graph.

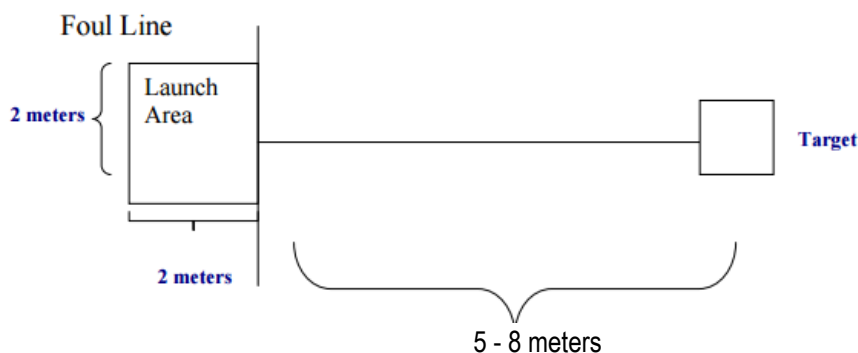
## **Impound:**

1. During the designated impound period, the team must impound their catapult/trebuchet, any materials needed to alter the device during the testing phase, a tennis ball, impact resistant safety goggles, and their design log. All items should be clearly marked with the team number.
2. During impound, Event Supervisors will evaluate all devices for safety and determine if the device is suitable for testing and meet the design requirements.

## **Launching:**

1. Once the team is called to set up for launching, no coaching or support from outside is allowed. This could result in the team being placed in Tier 3.

2. Impact-resistant safety goggles (or glasses) are required during launch time. If team members do not wear safety goggles while they are setting up and running their device, they will be given an initial warning and a penalty of +20 points. If a second warning is given, the team will immediately be placed in Tier 2.
3. Once teams enter the event area to compete, they may not leave the area or receive outside assistance, materials or communications until they are finished competing. A violation of this rule will result in the team being placed in Tier 3.
4. The distance from the “foul line” to the center of the target will be announced at the start of the competition. Students may place their catapult at any point behind the foul line up to 2 meters. See image below.



5. Total time once the team is called is 5 minutes. Each team will have a one-shot practice round after which they will shoot at the target two times. The second and third shots will count toward the official score.
6. The practice shot measurement will only be used in the case of a tie.
7. Teams must give ample warning to the supervisors prior to launch. Warning may be “3,2,1 launching...” etc. A penalty of +20 points will be assessed for failure to warn supervisors before each launch.
- 8.
9. Only successful launches will be measured. Launches are successful when a Tennis Ball's first bounce goes beyond the foul line. Only 2 shots are allowed, beyond the practice shot, regardless if they are successful or not. (An unsuccessful launch will be counted as 2m (200 cm) from the target for the purposes of averaging launches.)
10. The device may be moved by the participants within the 2-meter square launch area after each attempted launch. Teams may make adjustments to their device, between shots, while in the launch area using only materials that were impounded with the device. They may not change the device in any way that would violate impound rules or design parameters..
11. The distance from the target will be measured after each shot, including the practice shot, and announced to the team to allow them to make adjustments based on their data chart. **All measurements will be announced in centimeters.**
12. The distance from the center of the target will be measured to the tennis ball's initial point of impact as determined by the supervisor. Tennis balls landing outside of the sand pile area, less than 2 meters from the launch area, will also be measured according to the initial point of impact as determined by the supervisor. Tennis balls hitting the target on initial impact will receive a score of zero. Tennis balls with an initial impact beyond 2 meters will be scored as 2 meters.
13. No tennis balls should be shot before or during the catapult event except during the 5-minute specified practice round or actual competition.
14. Teams will be allowed to use their graph or any information from the design log during competition.
15. The event supervisor will make every attempt to advise students of any penalties incurred during the launch.

**Scoring:**

1. Low score wins.
2. Score is the average of the distance from the target in centimeters, measured to a single centimeter precision. 1 centimeter is 1 point
3. Devices which do not meet the design requirements will result in the team being placed in Tier 2.
4. Penalties may be assessed as listed in the rules.
5. Ties will be broken by the distance from the target on the third tiebreaker launch.

# Tennis Ball Catapult Scoring Rubric

Team #: \_\_\_\_\_

School/Team Name: \_\_\_\_\_

|                                                                                                |                    | Does the Catapult meet the requirements? (Y/N) |
|------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------|
| <b>IMPOUND :</b>                                                                               |                    |                                                |
| Catapult determined to be safe?                                                                | N= Cannot compete; |                                                |
| Catapult sits upon rubber or soft material so as to not damage floor?<br><b>Cannot compete</b> | N =                |                                                |
| Catapult is within design parameters.<br>Less than 50 cm tall and less than 50 ponds           | N = Tier 2         |                                                |
| Catapult labeled with team #?                                                                  | N= -Tier 2         |                                                |
| Tennis ball(s) impounded and labeled                                                           | N= Tier 2          |                                                |
| Design Log provided?                                                                           | N = Tier 2         |                                                |
| Impound on time?                                                                               | N = Tier 2         |                                                |
|                                                                                                | <b>TIER?</b>       |                                                |

## LAUNCH SCORE :

Points

|                                                                                                         |  |
|---------------------------------------------------------------------------------------------------------|--|
| Practice Launch - will be measured and announced to team and used only for breaking ties. (enter below) |  |
| Launch #1 distance from target _____ cm    failed launch = 200 cm                                       |  |
| Launch #2 distance from target _____ cm    failed launch = 200 cm                                       |  |
| Average of the two Launches=                                                                            |  |

## LAUNCH PENALTIES :

Penalty

|                                                             |  |
|-------------------------------------------------------------|--|
| Failure to announce launch (3,2,1 ...launching) + 20 points |  |
| Student or device crosses launch line? +20 points           |  |
| Tennis Ball supplied by Event Supervisor +20 points         |  |
| Goggles violation +20 points or Tier 2                      |  |
| Outside interference? Y= Tier 3                             |  |
| <b>TOTAL=</b>                                               |  |

PLEASE NOTE THAT THIS SCORESHEET IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES, THE RULES TAKE PRECEDENCE OVER THE SCORESHEET.

# WEATHER PERMITTING

**DESCRIPTION:** This event will test the team's knowledge of conducting investigations and using appropriate technology to build an understanding of weather.

## **COMPETITORS BRING:**

- Pencil/writing utensil
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

## **THE COMPETITION:**

1. The event may be run as a sit-down test, station or a combination of the two. Teams will answer questions and/or complete tasks to assess any of the following topics:
  - a. Water Cycle, including
    - i. Processes of evaporation, condensation, precipitation, and run off
    - ii. Types of precipitation
    - iii. Location of major sources of water and the water cycle
  - b. Clouds, including
    - i. Types of clouds - identification from images or descriptions. Cloud types are limited to cirrus, cirrostratus, cirrocumulus, altostratus, stratus, stratocumulus, nimbostratus, cumulus, mammatus, lenticular, fog, and contrails.
    - ii. Processes of cloud formation
    - iii. Cloud relationships to weather conditions
  - c. Identification and description of the function of weather instruments (thermometer, barometer, rain gauge, hygrometer, sling psychrometer, wind vane, anemometer, weather balloon, radar, satellite)
  - d. Weather forecasting - using weather data to identify, describe, and/or predict weather conditions. To include:
    - i. Description/definitions of weather-related terms (dew point, front, humidity, relative humidity, pressure, air mass, windspeed, etc.)
    - ii. Identification of weather station and weather map symbols
    - iii. Reading weather maps/weather station data
    - iv. Using weather maps/weather station data to describe expected weather conditions
    - v. Using weather maps/weather station data over time to predict weather
  - e. Severe Weather in Florida - Hurricanes
    - i. Storm categories
    - ii. Parts of a storm
    - iii. Cone of uncertainty, prediction models, etc
    - iv. Weather phenomenon that lead to hurricanes
  - f. Preparation for Severe weather
    - i. Understand the need for a preparedness plan
    - ii. Be aware of common precautions for severe weather; ie lightning, tornado warning, hurricane preparedness, etc.

**RELATED FLORIDA STANDARDS:**

- .SC.5.E.7.1 Create a model to explain the parts of the water cycle. Water can be a gas, a liquid, or a solid and can go back and forth from one state to another.
- SC.5.E.7.2 Recognize that the ocean is an integral part of the water cycle and is connected to all of Earth's water reservoirs via evaporation and precipitation processes.
- SC.5.E.7.3 Recognize how air temperature, barometric pressure, humidity, windspeed and direction, and determine the weather in a particular place and time.
- SC.5.E.7.4 Distinguish among the various forms of precipitation (rain, snow, sleet, and hail), making connections to the weather in a particular place and time.
- SC.5.E.7.7 Design a family preparedness plan for natural disasters and identify the reasons for having such a plan.

**SCORING:**

- High score wins.
- Points will be awarded for the accuracy of responses.
- Ties will be broken by the accuracy or quality of preselected questions chosen by the event supervisor.

# WRITE IT, DO IT

**DESCRIPTION:** Each participant will write a description of an object and how to build it and their partner will attempt to construct the object from this description.

**COMPETITORS BRING:**

- Pencil/writing utensil

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

**THE COMPETITION:**

1. The event will take place in two rooms.
2. During the first 20 minutes, the participants in each room will be shown an object, which may be abstract, but is the same for all teams. Example materials may be: drinking straws, paper cups, pipe cleaners, string, popsicle sticks, paper, etc.
  - a) Materials are not limited to this list.
  - b) The participant is not allowed to touch the object unless permitted by the Event Supervisors.
2. The participants have 20 minutes to write a detailed description of the object and how to build it.
  - a) Drawings and diagrams of the model are not allowed
  - b) Numerals, words, and single letters that fit within the content of the description are allowed
  - c) Participants may use abbreviations or punctuation that fit within the context of the description
3. The Event Supervisors will then change rooms. Students will receive their partner's instructions and attempt to recreate the original object in 20 minutes

**RELATED FLORIDA STANDARDS:**

- SC.4.N.1.8 Recognize that science involves creativity in designing experiments.
- SC.3.N.1.4 Recognize the importance of communication among scientists.
- SC.5.N.2.1 Recognize and explain that science is grounded in empirical observations that are testable; explanation must always be linked with evidence.
- SC.3.N.1.7 Explain that empirical evidence is information, such as observations or measurements that is used to help validate explanations of natural phenomena.
- SC.4.N.1.7 Recognize and explain that scientists base their explanations on evidence.
- SC.5.N.1.6 Recognize and explain the difference between personal opinion/interpretation and verified observation.

**SCORING:**

- High score wins
- Points will be awarded for proper size, color, location, orientation, and/or connection. The scores for both constructed objects will be added together to get the team's score.
- Pieces connected beyond an incorrect connection will be counted in the score. No penalty for unused parts.
- Students will be ranked Tier 2 if a subsection of the model is drawn.
- Drawing a complete picture of the model will result in disqualification.
- Ties will be broken by predetermined portions of the "build".

# SPECIAL EVENT: SHELBY JACOBS ROCKETRY

This special event will be run at the Jacksonville competition in memory of Shelby Jacobs (1935-2022). Mr. Jacobs was one of the first African American aerospace engineers and worked on rockets for the Mercury, Atlas, Jupiter, Thor, Apollo, and Space Shuttle programs. Mr. Jacobs was a huge fan of Florida Elementary Science Olympiad and the opportunities that it gives to students. The Florida First Coast Science Olympiad competition will host a water rockets competition in honor of Mr. Jacobs. This is a special event and will not count toward overall team standings. Winners will receive special awards provided by Mr. Jacobs's family.

**Description:** Prior to the competition, teams will design and build a rocket propelled by air pressure and water out of a 1-liter plastic beverage bottle. The students will create an Engineering Design Log (EDL) as they work through the design process and submit it to the event supervisor. The rocket that stays in the air for the longest time accompanied by the best EDL will win. This is a PRE-BUILT EVENT.

**Number of Participants:** 2

**Impound:** Yes

## **Construction:**

1. Teams will bring their rockets to the testing site in a box labeled with their team number. The rocket must also have the team number printed on it using a permanent marker.
2. The rocket fuselage/body must be made from a single clear, traditional plastic 1-liter beverage bottle used to hold water and air pressure that propels the rocket when released.
3. The 1-liter bottle must have a neck/nozzle opening internal diameter of approximately 2.2 cm. Some water bottles have different diameters and heights and they cannot be accommodated on the launcher. Some also have extended necks, making it impossible to attach to the launcher.
4. The bottle itself must not be altered in any way (e.g. holes, scratches, increasing the volume, restricting the bottle's opening). A safety check may be done to make sure that the bottle has not had any structural alterations due to physical, thermal, or chemical changes.
5. The rocket must not contain or be subject to any of the following: metal, commercial model rocket parts including the parachute, explosives, gases (except air), electronic devices, elastic, remote control, chemicals or pyrotechnics, super glue (cyanoacrylate), or high temperature hot glue.
6. Fins, parachutes, and other items may be added to the outside of the bottle to increase the time aloft. Adhesives that may be used to attach components to the bottle are limited to tape and glue (only silicone, polyurethane-based, or others that do not damage the structural integrity of the bottle).
7. All parts and any changes must be attached at a point that is 5 cm or more above the level of the bottle's opening. No part of the rocket can intrude into this 5 cm zone.
8. The nose of the rocket must be rounded or blunt with a maximum curvature that matches that of a penny. It can be less curved than a penny (i.e. not as sharp) but it can't be more curved (sharper) than a penny.
9. The fully assembled rocket must not exceed 50 cm in height and 20 cm in width, in its prelaunch state..
10. Energy to propel the rocket must come only from the water and air pressure in the bottle. Other sources of potential or kinetic energy are not allowed.
11. Only plain water may be used in the rocket. No other material of any type may be put in the bottle or added to the water. A water level line may be marked on the bottle to aid in adding water.
12. Parts of the rocket may deploy during flight, but they must remain attached by a string/lanyard.
13. Rockets will be launched from an Aquapod launcher. (available on online)

**Impound:**

1. Teams will impound their water rocket(s) in a box or container clearly marked with the team number. Event Supervisor during the impound process.
2. Teams will submit the Engineering Design Log to the Event Supervisor during impound.
3. Teams may impound repair kits containing tools, spare parts, and extra parachutes in their rocket box.
4. Any parts found to be dangerous (ex. glass or metal), illegal (ex. commercially made rocket parts), or that prevent a rocket from fitting on the launch pad must be removed before the rocket can be launched. Rockets that are changed to meet the construction requirements may be penalized.
5. A safety check may be done to make sure that the bottle has not had any structural alterations due to physical, thermal, or chemical changes.
6. Rockets deemed unsafe according to the event supervisor's judgment will not be launched.

### **The Competition:**

1. Two launches will be allowed. Different rockets may be used for each launch.
2. Teams must use the water, launch pad, and source of pressure provided by the event supervisors.
3. The team will add the desired amount of water to the rocket before each flight and may make alterations or repairs to rocket(s) between launches.
4. Outside assistance/coaching from the sidelines is not permitted and will result in the team being placed in Tier 2.
5. The event supervisors will supervise the pressurization of the rocket to 30 psi and the launching of it. Students must not begin any action without explicit direction of the event supervisor. Anyone within 5 meters of a pressurized rocket must wear eye protection. Contestants may not hold their rocket during pressurization.
6. One team member will pull the cord to launch the rocket. The student must call out a warning at launch such as "3,2,1 launch".
7. It will be the supervisors' decision whether the flight should be considered as unofficial due to variable weather conditions such as high winds or an error on the part of the Aquapod.
8. Timing begins when the rocket separates from the launcher.
9. Timing ends when any part of the rocket touches the ground, rests on an obstruction, or goes out of sight.

### **Engineering Design Log:**

Teams will be asked to submit an Engineering Design Log ([Engineering-Design-Log.pdf](#)):

1. Teams will use the template provided to create their EDL.
  - a. For "Identify a need" students should use the challenge set forth for this event.
  - b. "Specify Requirements" should clearly include design requirements for this event.
2. The team may make multiple copies of pages as needed to show the progression of their design(s).
3. The event supervisor will use the rubric provided in the EDL to help evaluate.
4. Teams will need to add a cover to the EDL with the team number and event title marked clearly.

### **Scoring:**

1. Highest score wins.
2. The times of both flights will be added together to achieve a total time score. This will be combined with the score from the Engineering Design Log.
3. Scores will be determined for the rocket by taking the average of the two event supervisors' times for the duration of each rocket's flight.
4. The watches will be started simultaneously at time of launch and stopped when the rocket touches the ground or any object in contact with the ground, trees, buildings, etc.

5. Flights of rockets whose parts do not remain attached together during the entire flight, or that cannot be changed to meet the construction requirements, will be ranked in Tier 2 and then by their time aloft, behind all flights of rockets without construction violations and whose parts remain attached.
6. Teams whose rockets cannot be launched for any reason will receive participation points only.
7. Ties will be broken using the quality of the Engineering Design Log.

Launcher Links:

[Oriental Trading Company](#)

[Alumigo](#)

[Altakorea](#)

# Special Event

## Shelby Jacobs Rocketry Scoresheet

Team Number: A\_\_\_\_\_

Team Name: \_\_\_\_\_

Student Names: \_\_\_\_\_

### IMPOUND

|                                                       |            |  |
|-------------------------------------------------------|------------|--|
| Were all items impounded during the appropriate time? | N = Tier 3 |  |
| Is the device within the design parameters?           | N= Tier 3  |  |

### LAUNCH

|                                                |           |  |
|------------------------------------------------|-----------|--|
| Did the students wear protective eye covering? | N= Tier 2 |  |
| Outside interference or coaching?              | Y=Tier 2  |  |

### SCORING

|                                                     |                   |
|-----------------------------------------------------|-------------------|
| Launch #1    Stopwatch 1: _____ Stopwatch 2: _____  | Average time:     |
| Launch #2    Stopwatch 1: _____ Stopwatch 2: _____  | Average time:     |
| Engineering Design Log Score (up to 20 points)      | Score:            |
| <b>Total of the two average flight times:</b> _____ | <b>Tier</b> _____ |

# SPECIAL EVENT: PROFESSOR JENSEN'S POTIONS

**DESCRIPTION:** This event will be run at the ORLANDO tournament. This event is in memory of Mrs. Samantha Jensen, a supporter of Florida Science Olympiad and a coach from Orlando Science Schools. Ms. Jensen's passing was a huge loss to the Florida Science Olympiad community. Special themed awards will be given for this event. This event does not count toward overall team standings.

Teams will be asked questions and/or will conduct experiments at stations as they relate to **science laboratory equipment, chemical reactions, toxins and antidotes. Safety goggles are required.**

## **COMPETITORS BRING/NEED:**

- Safety goggles
- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

**NUMBER OF PARTICIPANTS:** Maximum of 2

**IMPOUND:** NO

## **THE COMPETITION:**

1. This event leans heavily on elements from the [Harry Potter](#) series, a favorite of Ms. Jensen.
2. Each team will move from one station to another. There will be 5 stations that can be completed in any order even though they have numbers. Each station should take approximately the same amount of time.

### **Station 1: First Year Potions Class**

- a. This station focuses on activities related to lab safety and laboratory equipment such as:
  - i. What is the importance of PPE in the laboratory?
  - ii. What are standard safety expectations in a laboratory?
  - iii. What are the uses of common laboratory equipment, including graduated cylinders, beakers, flasks, hot water baths, bunsen burners, mortar and pestle, pipettes, pH paper, microscopes, test tubes, well plates, gloves, goggles, aprons, eye wash stations, etc..

### **Station 2: Second Year Potions Class**

- a. At Hogwarts, second-year students focused on a specific element of potions and brewed simple potions.
- b. This station focuses on simple chemical reactions that include Sodium Bicarbonate (baking soda), Acetic Acid (vinegar), Calcium Carbonate (egg shells), Calcium Chloride, water, and/or Ascorbic Acid (Vitamin C). Sample tasks may include:
  - i. Describing the physical and chemical properties of the compounds listed above.
  - ii. Performing simple experiments and collecting data using the compounds above
  - iii. Identifying the characteristics of a chemical change
  - iv. Writing simple chemical equations using the compounds listed above.

### **Station 3: Third Year Potions Class**

- a. At Hogwarts, third-year students focused on "Undetectable Poisons".
- b. This station focuses on toxic plants. Students will need to be able to identify these plants from images, explain the symptoms of poisoning from these plants, and describe possible remedies/antidotes. The plants that students may be tested on are:
  - i. Wolfsbane (*aconitum sp.*)
  - ii. Jack-in-the-pulpit (*Arum maculatum*)

- iii. Lily of the Valley (*Convallaria majalis*)
- iv. Poison sumac (*Toxicodendron vernix*)
- v. Poison Ivy (*Toxicodendron radicans*)
- vi. Water Hemlock (*Cicuta maculata*)
- vii. Tobacco (*Nicotiana tabacum*)
- viii. Mistletoe (*Viscum album*)

#### **Station 4: Fourth Year Potions Class**

- a. At Hogwarts, fourth-year students focused on “Poison Antidotes”
- b. This station focuses on toxic animals. Students will need to be able to identify these animals from images, explain the symptoms of poisoning/envenomation from these animals, and describe possible remedies/antidotes. The animals that students may be tested on are:
  - i. Poison dart frogs (*Dendrobates sp*)
  - ii. Portuguese man o’war (*Physalia physalis*)
  - iii. Fattail scorpion (*Androctonus australis*)
  - iv. Cone snails (*Class Conoidea*)
  - v. Cane toad (*Rhinella marina*)
  - vi. Box jellyfish (*Carukia barnesi*)
  - vii. Fire salamander (*Salamandra salamandra*)
  - viii. Olive Sea Snake (*Aipysurus laevis*)

#### **Station 5: Fifth Year Potions Class**

- a. At Hogwarts, fifth-year students focused on properly brewing potions using directions.
- b. Students will be asked to complete a lab activity following directions from the supervisor which may include:
  - i. Mixing reagents and measuring: reaction rate, production of gas, color change, development of a precipitate
  - ii. Separating mixtures
  - iii. Determination of pH

#### **RELATED FLORIDA STANDARDS:**

- **SC.5.P.8.1** Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature
- **SC.3.P.8.1** Measure and compare temperatures of various samples of solids and liquids.
- **SC.3.P.8.2** Measure and compare the mass and volume of solids and liquids.
- **SC.5.P.8.3** Demonstrate and explain that mixtures of solids can be separated based on observable properties of their parts such as particle size, shape, color, and magnetic attraction.
- **SC.5.P.8.2** Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process.
- **SC.5.P.9.1** Investigate and describe that many physical and chemical changes are affected by temperature.

#### **SCORING:**

1. **High score wins.**
2. Each question or station will be assigned a predetermined set of points, assigned by the event supervisor **and communicated to the teams (preferably- points are identified on the test/station paperwork).**
3. Ties will be broken by predetermined questions.

# COACH PAPERWORK

Coaches will generally have to bring the following paperwork to competition- the tournament director will share specifics once you have registered.

## Paperwork Required:

- Press and Website Release Form (for EVERY person- student, parent, coach, etc.)
- Student Code of Conduct form (signed by the coach and all competitors)
- Parent/Coach Code of Conduct form (signed by coaches and parents)
- Final Registration (MUST be signed by the school's principal/CEO)

## Arbitration Paperwork for use by the members of the team during the competition:

- Appeal Directions
- Appeal Form

**FLORIDA SCIENCE OLYMPIAD**  
**PRESS AND WEBSITE RELEASE FORM**

Date: \_\_\_\_\_

I \_\_\_\_\_

(Talent)

(Print name)

do hereby grant Florida Science Olympiad, INC (Producer) their affiliations, associations, exhibitors and distributors the full unrestricted right to the use of my picture, voice and likeness in multimedia presentations.

Talent also agrees to hold Producer Florida Science Olympiad, INC., their affiliations, association, exhibitors, distributors, properties and assets harmless in all litigations regarding this agreement and participation in the above mentioned program and its content.

Talent: \_\_\_\_\_

Signature

Parent or Guardian, for students under age 18: \_\_\_\_\_

Signature

Producer: Florida Science Olympiad, INC

# FLORIDA SCIENCE OLYMPIAD CODE OF CONDUCT

Dear Coaches:

Please direct students to read, to discuss, and to sign the following Code of Conduct. This document must be returned on the day of the competition.

Student participants are expected to compete in tournament events with an honest effort to follow the rules and spirit of the competition. The goal of the competition is to give one's best effort while displaying honesty, integrity, and sportsmanship.

Students, coaches, and parents are expected to display courtesy and respect toward Olympiad officials, other teams, and guests of the Olympiad. Please remember all officials are volunteer specialists in their field and should be extended every courtesy.

Failure to show honesty and/or courtesy by a participant, coach, or guest of the team may result in disqualification of the team from that event and/or from the entire competition.

## STUDENT SIGNATURES

|    |                 |
|----|-----------------|
| 1. | 9.              |
| 2. | 10.             |
| 3. | 11.             |
| 4. | 12.             |
| 5. | 13.             |
| 6. | 14.             |
| 7. | 15.             |
| 8. | 16. (Alternate) |

I have read and discussed the above document with my team \_\_\_\_\_  
(Coach's Signature).

Coach cell phone number for day of tournament: \_\_\_\_\_

# FLORIDA SCIENCE OLYMPIAD

## PARENTS AND TEACHERS

### CODE OF CONDUCT

Dear Coaches:

Science Olympiad is an event that encourages students to learn science and much is gained through success and failure. To that means, we encourage the students to participate according to the following code of conduct. However, that code also applies to all participants, including parents, chaperones and teachers. Adults are to encourage their children, but not interfere or communicate with them while the competition is underway. If judges determine that parents actively interfere or help their children during the tournament, the children may be disqualified from the event.

Please ask all adults who attend the competition to read and sign this document. This document must be returned on the day of the competition. "Student participants are expected to compete in tournament events with an honest effort to follow the rules and spirit of the competition. The goal of the competition is to give one's best effort while displaying honesty, integrity, and sportsmanship.

Students, coaches, and parents are expected to display courtesy and respect toward Olympiad officials, other teams, and guests of the Olympiad. Please remember all officials are volunteer specialists in their field and should be extended every courtesy. Failure to show honesty and/or courtesy by a participant, coach, or guest of the team may result in disqualification of the team from that event and/or from the entire competition."

| SIGNATURES |     |
|------------|-----|
| 1.         | 6.  |
| 2.         | 7.  |
| 3.         | 8.  |
| 4.         | 9.  |
| 5.         | 10. |

I have read and discussed the above document with my team \_\_\_\_\_.

Coaches Signature

## **INSTRUCTIONS FOR FILING AN ARBITRATION**

REMEMBER: EVENT JUDGES ARE EXTREMELY BUSY DURING THE COMPETITION. PLEASE DO NOT INTERFERE WITH AN EVENT WHILE IT IS IN PROGRESS.

1. MAKE AN ATTEMPT TO RESOLVE THE PROBLEM WITH THE EVENT SUPERVISOR (JUDGE) BEFORE SUBMITTING A WRITTEN ARBITRATION.
2. THIS FORM MUST BE TURNED IN TO TOURNAMENT HEADQUARTERS WITHIN 1 HOUR AFTER THE COMPLETION OF THE EVENT. IF CIRCUMSTANCES PREVENT FILING WITHIN THIS TIME LIMIT, THE ARBITRATION COMMITTEE MAY, AT THEIR DISCRETION, ACCEPT A LATE ARBITRATION.
3. PLEASE BE SPECIFIC IN YOUR APPEAL. BE SURE TO IDENTIFY THE EXACT RULE/PROCEDURE THAT YOU FEEL WAS NOT FOLLOWED. ARBITRATIONS THAT ARE GENERAL IN NATURE MAY BE REJECTED BY THE ARBITRATION COMMITTEE. THIS FORM MUST BE SIGNED BY THE OFFICIAL COACH OF THE TEAM.
4. THE EVENT SUPERVISOR (JUDGE) MUST SIGN THIS FORM BEFORE IT IS SUBMITTED TO THE ARBITRATION COMMITTEE. IF TIME PERMITS. HE/SHE MAY ENTER HIS/HER RESPONSE ON THIS FORM, OR ENTER THE TIME NOTIFIED AND RESPOND TO THE ARBITRATION COMMITTEE BEFORE THE FINAL SCORING FOR THE EVENT IS COMPLETED.
5. THE DECISION OF THE ARBITRATION COMMITTEE IS FINAL.

## FLORIDA SCIENCE OLYMPIAD ARBITRATION FORM

Event \_\_\_\_\_

Division \_\_\_\_\_ Coach \_\_\_\_\_

School \_\_\_\_\_ Number \_\_\_\_\_

What rule/procedure do you feel was not followed? (Cite specific page # and paragraph from coach's manual)

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Describe how rule/procedure was not followed (Attach additional sheets if necessary.)

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Suggested Resolution:

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Arbitrating Coach's Signature: \_\_\_\_\_

Event Supervisor's Opinion (Please do not interrupt judging)

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Event Supervisor's Signature: \_\_\_\_\_

Arbitrator's Resolution: (The decision of the arbitrators is final)

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Arbitrator's Signature: \_\_\_\_\_

**RETURN FORM TO HEADQUARTERS. DECISION POSTED AFTER 1 HOUR**

## FINAL REGISTRATION: SUBMIT THIS FORM AT REGISTRATION ON THE DAY OF THE EVENT.

On the day of the Florida Science Olympiad Tournament, coaches must submit a final registration form which identifies the team members. This list is used to ensure that teams are meeting organizational rules.

Team Name \_\_\_\_\_ Coach \_\_\_\_\_ Team Number \_\_\_\_\_

| Student Name | Grade | Student Signature |
|--------------|-------|-------------------|
| 1. _____     | _____ | _____             |
| 2. _____     | _____ | _____             |
| 3. _____     | _____ | _____             |
| 4. _____     | _____ | _____             |
| 5. _____     | _____ | _____             |
| 6. _____     | _____ | _____             |
| 7. _____     | _____ | _____             |
| 8. _____     | _____ | _____             |
| 9. _____     | _____ | _____             |
| 10. _____    | _____ | _____             |
| 11. _____    | _____ | _____             |
| 12. _____    | _____ | _____             |
| 13. _____    | _____ | _____             |
| 14. _____    | _____ | _____             |
| 15. _____    | _____ | _____             |

**I certify that all the above students are active members of our school and the grade levels are appropriately indicated.**

\_\_\_\_\_  
Principal's Signature

\_\_\_\_\_  
Coach's Signature